MORGAN PFAFF

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EDUCATION

University of California, Berkeley and San Francisco Berkeley, CA
Doctor of Philosophy in Bioengineering *Expected December 2026*

- GPA: 4.0/4.0
- National Science Foundation Graduate Research Fellowship Awardee
- Major area of study: Biomaterials Engineering and Tissue Regeneration
- Minor area of study: Cell, Tissue, and Developmental Biology
- Completed rotations in the laboratories of Dr. Kevin Healy, Dr. Phillip Messersmith, and Dr. Sanjay Kumar

Northeastern University Boston, MA
Bachelor of Science in Biochemistry *May 2021*

- GPA: 4.0/4.0
- Honors Program Member and Honors in the Discipline
- Coursework included Protein Chemistry, Cellular & Molecular Biology, Biochemistry, Biostatistics, Organic Chemistry, and Physical Chemistry, as well as a general survey of Computer Science, Physics, and Calculus
- Participated in an Engineering study abroad program in Oxford, United Kingdom, Summer 2019

RESEARCH EXPERIENCE

Kevin Healy Lab, UC Berkeley/UCSF Berkeley, CA
PhD Candidate *January 2022 – Present*

- Engineer hyaluronic acid-based hydrogels for volumetric muscle loss and delayed rotator cuff repair; co-inventor on MuscleMatrix, an early-stage therapeutic device built on this research
- Work has resulted in 2 peer-reviewed publications and 4 conference presentations to date

Directed Assembly of Particles and Suspensions, Northeastern University Boston, MA
Biomaterials Undergraduate Researcher *September 2016 – July 2021*

- Wrote a grant proposal for and received the prestigious Project-Based Exploration for the Advancement of Knowledge Summit Award to fund my research
- Developed a protocol to fabricate dual-network hydrogels with magnetically responsive hydroxyapatite fiber reinforcement for bone graft applications and tailored composite design to maximize toughness
- Characterized and investigated the mechanical strength of hydrogel samples by independently conducting sensitive experiments using scanning electron microscopy and dynamic mechanical analysis
- Designed, ran, and analyzed simulations using COMSOL Multiphysics FEA software of hydroxyapatite fiber rotation in an alginate matrix to investigate a potential new toughening mechanism
- Manufactured freeze-casted conduits with highly aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School
- Work resulted in an oral presentation and 4 poster presentations

Vertex Pharmaceuticals Boston, MA
Exploratory Research Co-op *January 2020 – June 2020*

- Developed and optimized cellular assays for transcription factor detection in potential druggable pathways for the treatment of Sickle Cell Disease (SCD)
- Conducted pilot experiments to determine expression profiles of various proteins over time

- Utilized molecular cloning of recombinant vectors for stable integration into cell lines of interest
- Independently planned, executed, and analyzed results from experiments and regularly communicated findings with other members of the Exploratory Research group
- Analyzed and presented on relevant articles during Journal Clubs in front of the SCD team

Editas Medicine Cambridge, MA
Screening Group Co-op *January 2018 – June 2018*

- Screened for the most effective sgRNAs for the company's pipeline diseases and performed CRISPR-mediated mutations of genes of interest via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays for automated liquid handlers
- Conducted a complete comparative analysis of the company's Biomek i7 automated liquid handler and its Biomek FX predecessor
- Utilized software such as Prism, Tableau, and FlowJo to visualize, analyze, and compare data, and presented findings to other members of the screening group and other departments

University of Connecticut Health Center Farmington, CT
Summer Research Fellow *June 2017 – August 2018*

- Piloted synthetic-lethal experiments to find exploitable vulnerabilities in cell signaling pathways and elucidate novel druggable osteosarcoma targets
- Tested the effects of various inhibitor and siRNA combinations on undifferentiated hFOB cells and osteosarcoma cell lines SaOS2 and U2OS to specifically target tumor cells with mutations interfering with growth factor receptor regulation
- Practiced sterile technique to grow and maintain cell cultures for several months and completed analyses using PCR and absorbance assays
- Work resulted in a poster presentation and provided the foundation for NIH grant proposals from this lab

RESEARCH SKILLS

Cellular & Molecular Biology
Aseptic Technique, Automated Liquid Handling (Biomek i7, Biomek FX, Mantis, Tempest), ddPCR, DNA Extraction, DNA Purification, Flow Cytometry, Gel Electrophoresis, High-Powered Cell Disruption, Laser Cell Capture, Lipofection, Mammalian Cell Culture, Molecular Cloning, Nucleofection, qPCR, Restriction Digest, Viral Transduction, Western Blot

Material Fabrication / Characterization
Dynamic Mechanical Analysis, Freeze Casting, Lyophilization, Nuclear Magnetic Resonance (NMR) Spectroscopy, Optical Microscopy, Scanning Electron Microscopy, Sonication, Tensile Testing

Programming
Proficient in LaTeX, Java, Microsoft Office (Word, Excel, PowerPoint, Outlook); familiar with C++, MATLAB, Arduino

Statistical Analysis, Graphing, and Simulation Software
R, Tableau, GraphPad Prism, FlowJo, SnapGene, COMSOL Multiphysics

ADDITIONAL WORK EXPERIENCE

Tufts Medical Center, Oncology and Hematology Boston, MA
Clinical Care Technician *January 2019 – December 2019*

- Assisted 6–10 patients per shift with tasks of living such as ambulation, bathing, dressing, and eating
- Monitored patient vital signs (blood pressure, temperature, pulse, O2 saturation), blood sugar levels, and overall patient condition, and reported abnormal conditions to the assigned nurse
- Supported nurses and physicians during examinations, medical procedures, and emergency situations
- Responded to the needs of patients and family members and took a proactive role in the efficient operation of the hospital

Northeastern University Housing and Residential Life Boston, MA
Resident Assistant *August 2017 – July 2019*

- Maintained a safe and orderly environment in the freshman residence hall to ensure the well-being of 45 female residents
- Confronted and resolved issues surrounding diversity, racism, alcohol and drug abuse, academic performance, and social acceptance in the university setting
- Individually and collaboratively created, planned, and implemented educational and recreational programs intended to facilitate the transition of incoming students to college

Fallon Ambulance Service Quincy, MA
Emergency Medical Technician – Basic *September 2018 – April 2019*

- Conducted medical and psychological assessments of ill or injured patients, provided appropriate immediate care, and safely transported them to the appropriate hospital or specialized medical facility
- Interacted with patients and families during transports and emergencies and maintained a calm, safe environment for all parties
- Collected pertinent information including vitals, symptoms, medical history, allergies, and prescriptions and recorded it according to company protocols

Dental Office of Dr. Andrew J. Ponichtera, DMD, MS Simsbury, CT
Dental Assistant *June 2016 – August 2016*

- Assisted the dentist with procedures including fillings, root canals, and insertion of crowns, implants, bridges, and dentures
- Handled front desk operations including greeting patients, booking and confirming appointments, and organizing patient files
- Took inventory of dental supplies and equipment and placed new orders as needed

PUBLICATIONS & PRESENTATIONS

Publications

Viscoelastic HyA Hydrogel Promotes Recovery of Muscle Quality and Vascularization in a Murine Model of Delayed Rotator Cuff Repair

M. R. Pfaff, A. Wague, M. Davies, A. R. Killaars, D. Ning, S. Garcia, A. Nguyen, P. Nuthalapati, M. Liu, X. Liu, B. T. Feeley, K. E. Healy

Advanced Healthcare Materials, 2025, 14, 2403962 · doi.org/10.1002/adhm.202403962

NIPAAm-co-PEG4000 Thermoresponsive Hydrogel with ROS Scavenging Properties Improves In Vivo Border Zone Contractility and Reduces Myocardial Remodeling in Sheep After MI

V. Y. Wang, Y. Zhu, K. A. Spaulding, M. Pfaff, G. Neiman, K. Takaba, N. Chen, K. Parekh, A. Hasaballa, H. Haraldsson, A. Baker, D. Saloner, A. W. Wallace, N. P. Ziats, K. E. Healy, M. B. Ratcliffe

Acta Biomaterialia, 2026, 211, 48–60 · doi.org/10.1016/j.actbio.2025.06.046

Oral Presentations

Viscoelastic HyA Hydrogel Promotes Recovery of Muscle Quality and Vascularization in a Murine Model of Delayed Rotator Cuff Repair

Morgan R. Pfaff, Aboubacar Wague, Michael Davies, Anouk R. Killaars, Derek Ning, Steven Garcia, Anthony Nguyen, Prashant Nuthalapati, Mengyao Liu, Xuhui Liu, Brian T. Feeley, Kevin E. Healy

Biological Scaffolds for Regenerative Medicine (BSRM) · May 2025 · Napa, CA

Tunable Semi-Synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration After Rotator Cuff Injury

Morgan Pfaff, Anouk Killaars, Derek Ning, Michael Davies, Anthony Nguyen, Prashant Nuthalapati, Mengyao Liu, Xuhui Liu, Brian Feeley, Kevin E. Healy

Biomedical Engineering Society (BMES) Annual Meeting · October 11–14, 2023 · Seattle, WA

Semi-Synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration by Inducing BAT Differentiation of FAPs

Morgan Pfaff, Anouk Killaars, Derek Ning, Michael Davies, Anthony Nguyen, Prashant Nuthalapati, Mengyao Liu, Xuhui Liu, Brian Feeley, Kevin Healy

Society For Biomaterials Annual Meeting & Exposition, “Biomaterials for Regenerative Engineering” · April 21, 2023 · San Diego, CA

Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated

Pfaff, M., Faust, J., Erb, R.

Northeastern Regional Student Conference of the Society for Experimental Mechanics · June 22, 2020 · Dartmouth, MA

Poster Presentations

Tunable Semi-synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration After Rotator Cuff Injury

Morgan Pfaff, Anouk R. Killaars, Derek Ning, Michael Davies, Anthony Nguyen, Prashant Nuthalapati, Mengyao Liu, Xuhui Liu, Brian T. Feeley, Kevin E. Healy

ORS 50th International Musculoskeletal Biology Workshop · July 22–27, 2023 · Midway, UT

Semi-synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration by Inducing BAT Differentiation of FAPs

Morgan Pfaff, Anouk Killaars, Derek Ning, Michael Davies, Anthony Nguyen, Prashant Nuthalapati, Mengyao Liu, Xuhui Liu, Brian Feeley, Kevin Healy

Society For Biomaterials (SFB) Annual Meeting & Exposition · April 2023 · San Diego, CA

Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism

Pfaff, M., Erb, R.

Research, Innovation, and Scholarship Expo · April 15, 2021 · Boston, MA

Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated

Pfaff, M., Faust, J., Erb, R.

Materials Research Society 2020 Spring/Fall Meeting and Exhibit · November 28, 2020 · Boston, MA

Synthetic Lethal Approaches to Treating Osteosarcoma

Pfaff, M., Alander, C., Hansen, M.

University of Connecticut Health Center Summer Research Fellowship Program · July 27, 2017 · Farmington, CT

Dual-Network Hydrogels with Functionalized CaP Rods for Bone Graft Applications

Pfaff, M., Faust, J., Erb, R.

Research, Innovation, and Scholarship Expo · April 13, 2017 · Boston, MA

Dual-Network Hydrogels with Functionalized CaP Rods for Bone Graft Applications

Pfaff, M., Faust, J., Erb, R.

National Conference on Undergraduate Research · April 6–8, 2017 · Memphis, TN

FELLOWSHIPS, AWARDS, AND HONORS

Fellowships, Grants, and Scholarships

- **National Science Foundation Graduate Research Fellowship** (2021) — awarded to outstanding graduate students pursuing research-based master's and doctoral degrees in fields within NSF's mission ($138,000)
- **Project-Based Exploration for the Advancement of Knowledge “Shout It Out!” Award** (2020) — conference and travel grant for undergraduate researchers accepted to present at an external venue or conference
- **Project-Based Exploration for the Advancement of Knowledge Summit Award** (2020) — research grant awarded to undergraduate researchers with the skills to work independently, formulate creative research questions, and pursue them ($3,000)
- **Northeastern University Conference Travel Fund Award** (2017) — travel grant for students presenting original research at professional conferences
- **Presidential Global Scholarship** (2016) — awarded to outstanding students committed to academic success and global experience ($6,000)
- **Northeastern University Connections Merit Scholarship** (2016) — merit scholarship for female students entering STEM in the top 10–15% of the applicant pool ($18,000/year)

Awards and Honors

- **RISE Excellence in Research Award** (2021) — awarded to 1 of over 450 presenters for methodological and procedural excellence at the Northeastern University Research, Innovation, and Scholarship Expo ($2,000)
- **Northeastern University Huntington 100** (2021) — awarded to 100 outstanding students each year
- **Sigma Xi Scientific Research Honor Society** (2020–2021) — membership awarded for independent investigation resulting in new scientific discoveries
- **Northeastern University President's Award** (2020–2021) — awarded to the top 10 students of each university class by GPA
- **Northeastern University Dean's List** (2016–2021) — top 10–15% of students in the College of Science each semester
- **RISE Innovation Alley** (2017) — poster selected for the highest-impact project corridor at the Research, Innovation, and Scholarship Expo

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MENTORING & COMMUNITY OUTREACH

Food Not Bombs, East Bay — Oakland, CA
Volunteer — *September 2024 – Present*

- Serve 25–40 meals a week to the community as part of a weekly mutual-aid effort

Adventure Challenge — Boston, MA
Volunteer — *September 2018 – December 2019*

- Planned activities for and accompanied my Adventure Challenge partner, a 13-year-old girl with autism, to the YMCA pool to swim and play pool games on a weekly basis
- Promoted physical activity, social interaction, and overall health and wellness to help her maintain a healthy lifestyle

Northeastern University Housing and Residential Life — Boston, MA
Resident Assistant — *August 2017 – July 2019*

- Served as a mentor for 45 freshman women from around the world pursuing careers in STEM
- Helped residents set and achieve short- and long-term goals, network with established scientists and professors, and access resources to fulfill their academic and scientific aspirations
- Designed programs including Q&A panels with graduate students and professors, leadership and team-building workshops, and networking events with faculty mentors

Advancement Via Individual Determination (AVID) — Boston, MA
Tutor — *September 2018 – December 2018*

- Served as a discussion mediator during highly interactive tutoring sessions for students in the AVID program
- Assisted students in learning how to guide each other toward solutions via questions and critical thinking to develop intellectually and build effective, long-term study habits
- Promoted skills to help students pursue and succeed in higher education after graduating high school

University of Connecticut Health Center Farmington, CT
Volunteer *June 2018 – August 2018*

- Greeted and seated incoming patients arriving for injections or surgery at the Musculoskeletal Institute
- Alleviated pressure on front desk staff by assembling and organizing patient charts and answering phone calls

CERTIFICATIONS

- **U.S. Center for SafeSport Certification** (2019) — training on mandatory reporting, sexual misconduct awareness, and emotional and physical misconduct among youth and student athletes
- **Emergency Medical Technician, Basic** (2018) — certified EMT-B in Massachusetts, Connecticut, and nationally

ORGANIZATIONS & GROUPS

- **Sigma Xi, The Scientific Research Honor Society** (March 2020 – Present) — 2021–Present: Berkeley Chapter Vice President; Chairperson of the Committee for Student Advancement, a committee I developed to enhance student representation and success in the Berkeley Chapter. 2020–2021: Diversity & Inclusion Chair and Associate Member of the Northeastern University Chapter
- **Northeastern University Club Sports Council** (September 2019 – Present) — served as liaison between Club Sports teams and Club Sports Administrators and Directors
- **Northeastern University Triathlon Team** (September 2016 – Present) — President/Captain (2019–2020), Chief Financial Officer (2018–2019), Chief Procurement Officer (2017–2018)